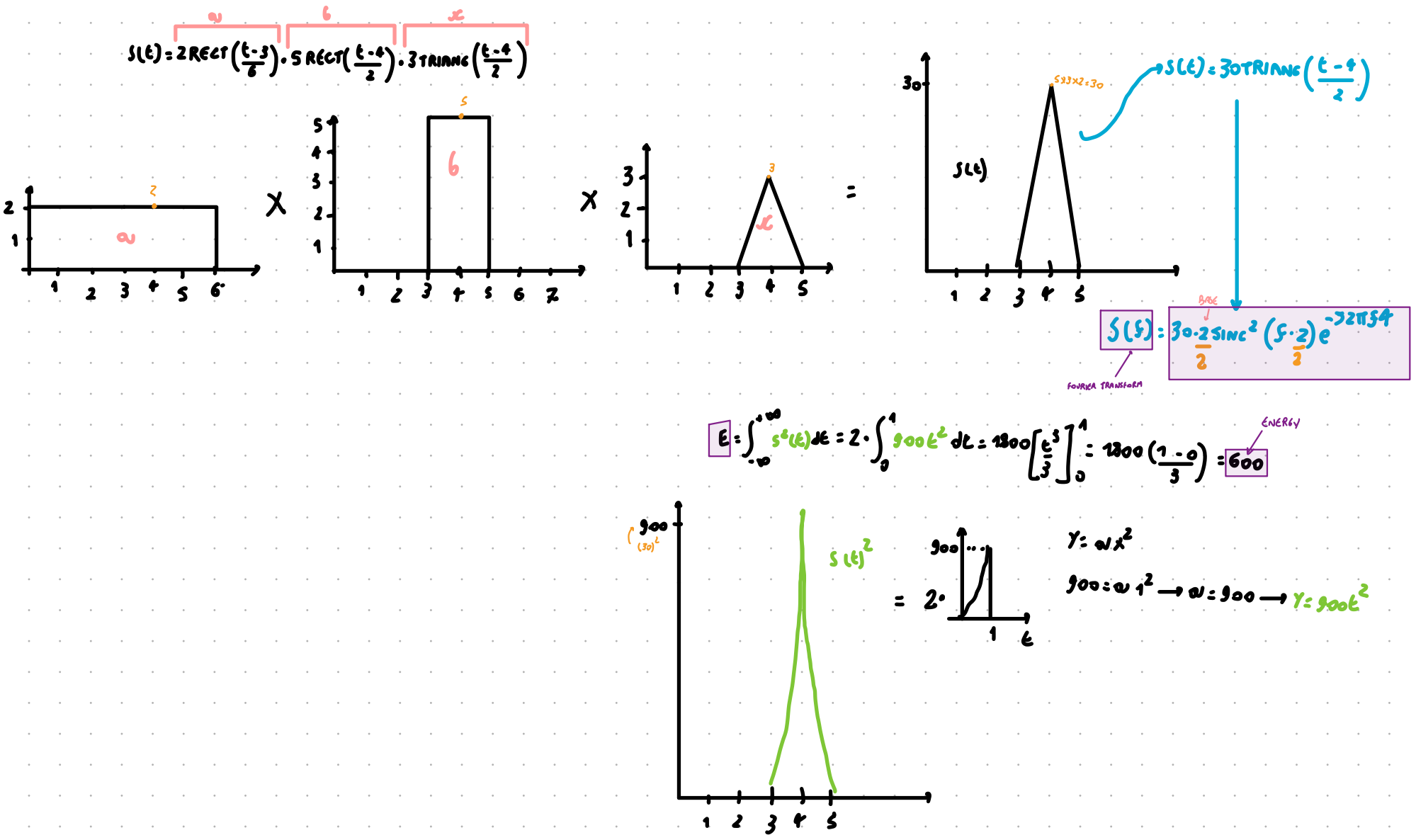


Exercise 4

It is given the signal $s(t) = 2rect\left(\frac{t-3}{6}\right) \cdot 5rect\left(\frac{t-4}{2}\right) \cdot 3triang\left(\frac{t-4}{2}\right)$, where the values are expressed in volts and the times in milliseconds. Draw it and determine its Fourier transform and energy.

Replicate infinite times the part of s(t) included in the interval $0 < t < 6$ thus obtaining a periodic signal r(t). What would be the value of the component at the fundamental frequency of the Fourier transform of r(t)?



PERIODIC SIGNAL

